

Geospatial Data Analysis with Python

Groundwater Quality and Arsenic Analysis in Bangladesh

Introduction

This project demonstrates how geospatial data can be processed, analysed, and visualized using Python. My main question was **How can groundwater sampling data with latitude and longitude be transformed into geospatial data and used to investigate the distribution of arsenic across Bangladesh?** Groundwater sampling records containing latitude and longitude coordinates were transformed into spatial point data using GeoPandas. These points were then spatially joined with Bangladesh administrative district boundaries to associate each groundwater sample with its geographical district. The new Python library used for this project is **GeoPandas** along with Pandas and Matplotlib.

Purpose

The purpose of this project is to investigate how Python can be used to analyse groundwater quality data together with geographical information. The main focus is on understanding how groundwater sampling data containing latitude and longitude coordinates can be transformed into geospatial data and then used to investigate the geographical distribution of arsenic concentrations across Bangladesh. Another purpose is to gain practical experience with GeoPandas, a Python library for working with geospatial data. I wanted to understand how GeoPandas can be used to create spatial point data, work with coordinate reference systems (CRS), perform spatial joins, combine different geographical datasets, save the processed geospatial data in a reusable format and produce maps for analysis.

My chosen field and its relevance to Data science

My chosen field is Geospatial Data Analysis with Python(GeoPandas), with a focus on Groundwater Quality and Arsenic Analysis in Bangladesh. Geospatial data analysis combines geographical information with traditional data analysis. It is a part of Environmental Data Science. Data scientists often need to work with location-based data, combine datasets from different sources, identify geographical patterns, and communicate findings through visualizations. In this project, I use GeoPandas, together with Pandas and Matplotlib, to transform groundwater sampling coordinates into geospatial data and analyse how arsenic concentrations are distributed across different geographical areas of Bangladesh.

Key concepts

Creation of the GeoDataFrame

After cleaning the coordinate data, the latitude and longitude values were converted into point geometries. GeoPandas was used to create the GeoDataFrame. Each groundwater record therefore became a geographic point. For example, the geometry is represented in the form POINT (90.7844 22.8731) where longitude is the X coordinate and latitude is the Y coordinate. The resulting GeoDataFrame contained 3,534 groundwater points.

Coordinate Reference System(CRS)

The groundwater coordinates were stored as geographic latitude/longitude coordinates. The project therefore uses EPSG:4326. EPSG:4326 is the WGS 84 geographic coordinate reference system. The boundary datasets were also loaded using the same coordinate system, allowing the spatial layers to be used together without requiring an additional CRS transformation.

Spatial Join

One of the most important parts of the project was assigning each groundwater sampling point to its geographical district. Instead of relying only on the district name already stored in the CSV, a genuine geospatial spatial join was implemented using GeoPandas. This is an important geospatial operation because the district assignment is derived from the actual geographic relationship between the sampling location and the administrative boundary.

Spatial aggregation

After assigning groundwater points to districts, I could group the observations geographically and calculate statistics such as median arsenic concentration.

This allowed me to move from individual sampling points to division-level and district-level analysis.

GeoDataFrame.explore(): This is a GeoPandas method for creating interactive maps from spatial data. It returns an interactive map that can be viewed in a web browser, allowing users to zoom, pan, and inspect individual geographic features.

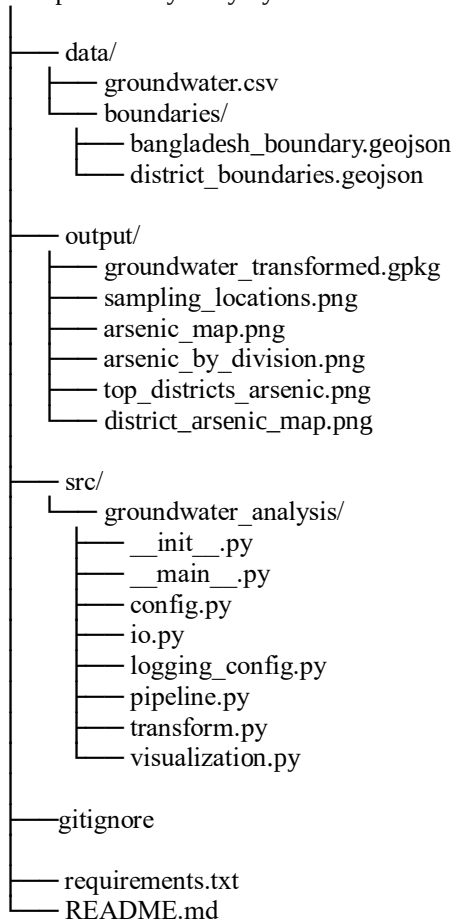
Project Overview

- ⑩ Load and clean groundwater sampling data.
- ⑩ Convert latitude and longitude coordinates into geographic point geometries.
- ⑩ Create a GeoDataFrame using GeoPandas.
- ⑩ Assign an appropriate Coordinate Reference System (CRS).
- ⑩ Load Bangladesh administrative boundary data.
- ⑩ Perform a spatial join between groundwater points and district boundaries.
- ⑩ Analyse arsenic concentrations geographically.
- ⑩ Create informative geographic visualizations.
- ⑩ Save the processed geospatial data in a reusable GeoPackage format.
- ⑩ Organize the project into separate modules with clear responsibilities.

Project Structure

The project was organized into separate directories for data, output, source code, and configuration.

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The separation of the source code into modules makes the project easier to understand and maintain.

Data

I have used groundwater dataset CSV file (NationalSurveyData.csv, British Geological Survey / DPHE Bangladesh groundwater data), Bangladesh administrative boundary shape files (national boundary and district-level boundaries). The national boundary contains the overall geographical boundary of Bangladesh. The district boundary dataset contains 64 district boundary features. The district layer contains both Polygon and MultiPolygon geometries. The district name field used for the spatial analysis was adm2_en.

After cleaning of groundwater dataset there are 3534 groundwater sampling records. The arsenic concentration column contained both exact measurements and censored measurements(<0.5, <6). To allow numerical analysis, the < symbol was removed and the remaining value was converted to numeric form. A second column, As_censored, was created to preserve information about whether the original observation was censored or not. This approach allows the project to perform numerical analysis while still retaining information about censored measurements. There are 3,534 total arsenic observations where 2429 exact numeric observations and 1105 censored observations.

Modular Code Design

The project was divided into several Python modules rather than keeping the entire program in one file. config.py contains project paths and output filenames. This keeps file locations centralized and makes the project easier to modify. io.py responsible for loading groundwater data, boundary layers and saving the final GeoDataFrame. transform.py responsible for cleaning groundwater data,

transforming arsenic values, creating the GeoDataFrame, performing the spatial join. `visualization.py` contains the visualization functions for sampling locations, arsenic concentration, division comparison, top districts and district-level map. `pipeline.py` controls the complete workflow and connects the different modules. `__main__.py` allows the project to be executed as a Python module by running `python -m src.groundwater_analysis` in terminal. This provides a clean entry point for running the analysis.

Project Pipeline

The complete project is organized as a reproducible pipeline. The workflow can be summarized as:

groundwater.csv → Load groundwater data → Clean data → Prepare arsenic values → Create GeoDataFrame → Set CRS(EPSG:4326 / WGS84) → Load district boundaries → Spatial Join → Spatial district match / Original district fallback → `district_final` → Save GeoPackage → Sampling map / Arsenic map / Division analysis / District-level analysis

Logging

The project uses logging rather than relying entirely on ordinary `print()` statements during pipeline execution. Logging makes it easier to monitor the execution of the analysis and diagnose problems without mixing debugging output with the actual analysis results.

Handling Spatial Join Mismatches

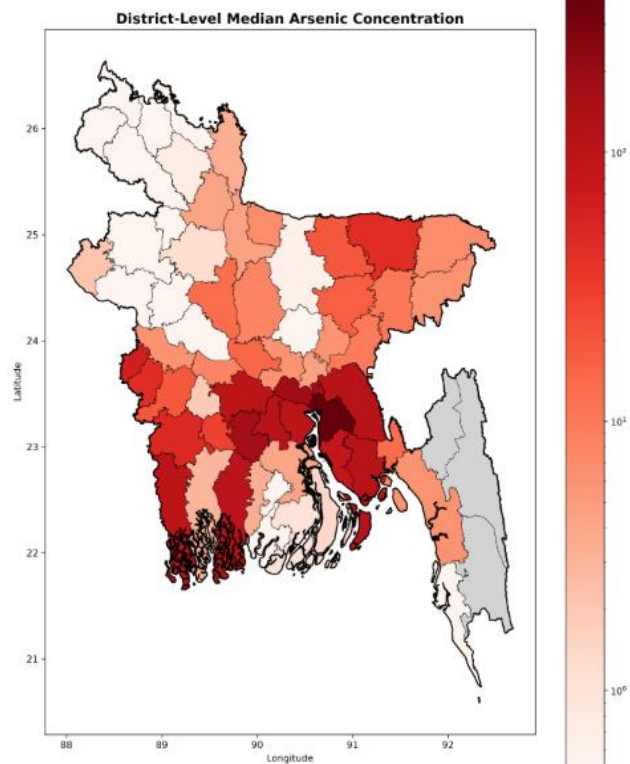
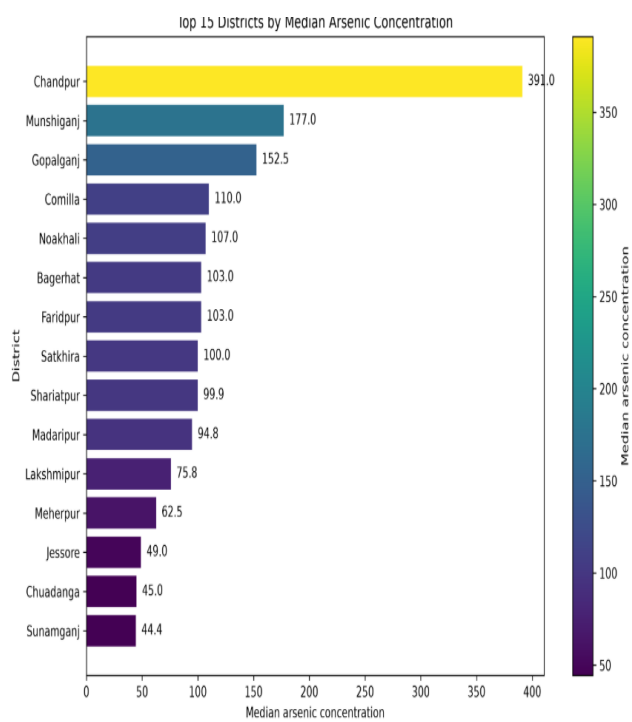
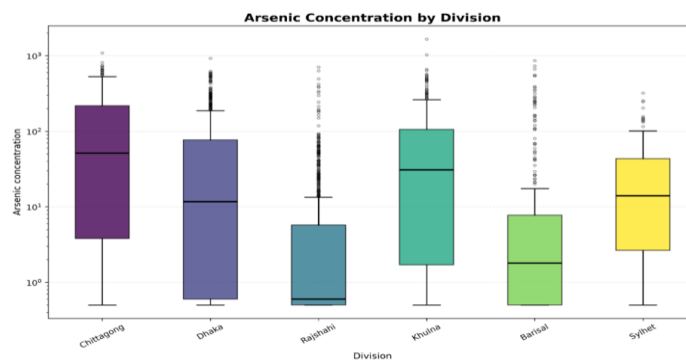
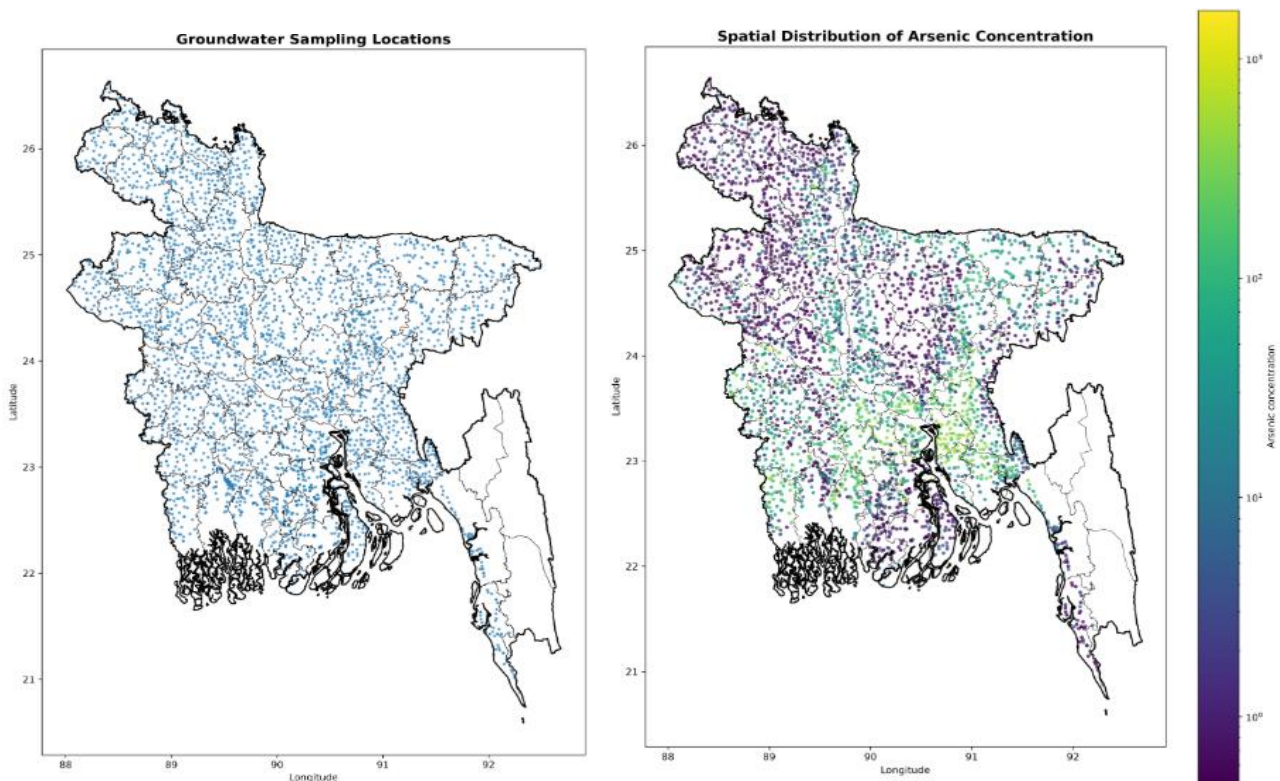
The spatial join successfully matched most groundwater points to district polygons. The results were 3,518 points spatially matched but 16 points did not receive a spatial district match.

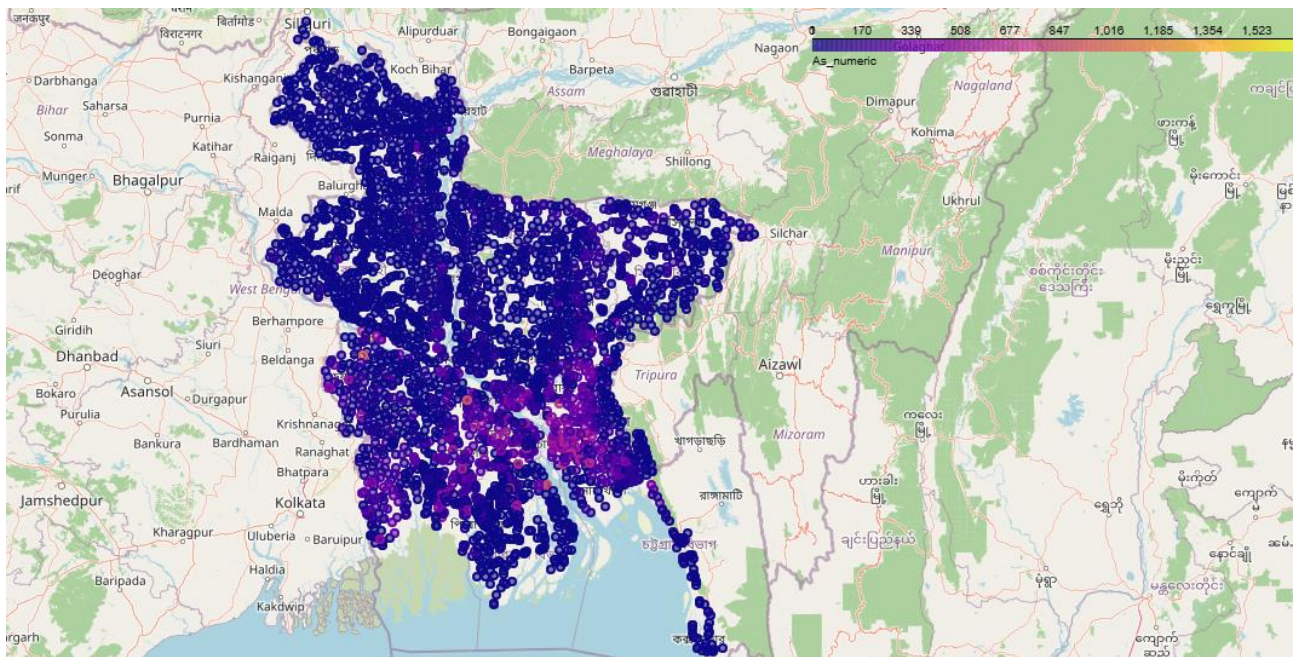
The 16 unmatched points were not simply deleted. Instead, the project retains the original `DISTRICT` value from the groundwater dataset as a fallback. Two fields were created: `district_spatial` and `district_final`. `district_spatial` contains the district obtained from the spatial join. `district_final` uses the spatial result where available and falls back to the original groundwater district value when necessary. This resulted in 3,534 records with a final district assignment. This approach preserves all groundwater observations while clearly distinguishing the spatially derived information from the original attribute information.

Output Files

The completed pipeline generated six output files: `groundwater_transformed.gpkg` (The processed spatial groundwater dataset), `sampling_locations.png` (Map of groundwater sampling locations), `arsenic_map.png` (Geographical visualization of arsenic concentrations), `arsenic_by_division.png` (Comparison of arsenic distributions across administrative divisions), `top_districts_arsenic.png` (Visualization of districts with high median arsenic concentrations) and `district_arsenic_map.png` (District-level choropleth map showing median arsenic concentration). `GeoDataFrame.explore()` returns an interactive map `arsenic_interactive_map.html`. All six outputs were successfully generated by the final pipeline.

GeoPackage was selected because it is a standard, portable format for storing geospatial vector data. The GeoPackage can therefore be reused for further analysis without repeating the entire data preparation process.





GeoDataFrame.explore() returns an interactive map

Groundwater Sampling Location Map: The first visualization shows the geographical distribution of groundwater sampling locations(sampling_locations.png). The groundwater samples are displayed as points over the Bangladesh administrative boundary.

Arsenic Concentration Map: A second visualization was created to display the geographical distribution of arsenic concentration (arsenic_map.png). The groundwater sampling points are represented geographically and coloured according to arsenic concentration.

Arsenic Analysis by Division and District-Level with Visualization: The project also analyses arsenic concentration at the administrative division level. The following six divisions are represented in the groundwater dataset: Dhaka, Chittagong, Rajshahi, Khulna, Barisal and Sylhet. A boxplot was created to visualize the distribution of arsenic concentrations between divisions(arsenic_by_division.png). The project also calculates median arsenic concentration at the district level. A separate visualization was created to highlight districts with high median arsenic concentrations(top_districts_arsenic.png). The project also produces a district-level choropleth map(district_arsenic_map.png). The district polygons are coloured according to median arsenic concentration. A logarithmic colour normalization was used because arsenic concentrations have a large range. The map also identifies districts without groundwater samples using a separate missing-data appearance. This distinction is important because a district without samples cannot be interpreted as having low arsenic concentration.

Interactive Arsenic Map: GeoDataFrame.explore() method from GeoPandas used for creating interactive map that can be viewed in a web browser, allowing users to inspect individual geographic features. In this project, explore() is used to display groundwater sampling locations and visualise arsenic concentrations using a colour scale.

Limitations

The analysis has several limitations that should be considered when interpreting the results.

First, the arsenic dataset contains censored measurements. In this project, values such as <0.5 were converted to 0.5 for numerical analysis. This is convenient for exploratory analysis but does not fully represent the statistical uncertainty associated with censored observations. Second, sixteen points did not spatially intersect the district boundary polygons. Rather than discarding these observations, their original district attributes were used as a documented fallback.

Finally, a high district median in this dataset should not automatically be interpreted as proof that the entire district has high groundwater arsenic levels. Sampling coverage, sample size, well characteristics, and sampling methodology should also be considered.

Conclusion

This project demonstrates a complete geospatial data-analysis workflow using Python and GeoPandas. Starting with raw groundwater CSV data, the project performs data cleaning, arsenic-value preparation, coordinate conversion, GeoDataFrame creation, coordinate reference system assignment, administrative boundary integration, spatial joining, statistical analysis, and geographic visualization. The final dataset contains 3,534 groundwater sampling records. The spatial join successfully assigned 3,518 records directly from district polygons, while 16 records required the original district information as a fallback, resulting in a final district assignment for all records. The project produces both a reusable GeoPackage and multiple visualizations that allow groundwater sampling patterns and arsenic concentration distributions to be examined geographically. Overall, the project demonstrates how Python-based geospatial techniques can transform ordinary tabular environmental data into a spatial dataset suitable for geographic analysis and visualization.

Relevance to the professional role

Geospatial data analysis can be a useful skill for a Data Scientist when working with data where **location is an important factor**. Data Scientist may need to combine geographical data with other datasets to find observations with geographical areas, and analyse patterns across different regions.

The skills I developed in this project, particularly using **GeoPandas for spatial data processing and spatial joins**, can be applied to areas such as environmental monitoring, public health, agriculture, infrastructure, and urban planning. This project also demonstrates how my existing knowledge of **Pandas, Python and Matplotlib** can be extended to work with spatial data. This gives me an additional data-analysis skill that can be applied when geographical information is an important part of a real-world dataset.

References:

1. GeoPandas, 'User Guide', *GeoPandas Documentation*, https://geopandas.org/en/latest/docs/user_guide.html, accessed 11 September 2026.
2. British Geological Survey, 'DPHE/BGS National Hydrochemical Survey' [data set], <https://www.bgs.ac.uk/groundwater/download/dphe-bgs-national-hydrochemical-survey/>, accessed 19 September 2026.
3. Directorate General of Health Services, 'Bangladesh Administrative Boundary – National Level (ADM0)' [geospatial data], DGHS GIS Portal, https://gis.dghs.gov.bd/server/rest/services/Hosted/bgd_admbnda_adm0_bbs_20201113/FeatureServer/0, accessed 19 September 2026.
4. Directorate General of Health Services, 'Bangladesh Administrative Boundaries – District Level (ADM2)' [geospatial data], DGHS GIS Portal, https://gis.dghs.gov.bd/server/rest/services/Hosted/bgd_admbnda_adm2_bbs_20201113/FeatureServer/0, accessed 19 September 2026.

Self-reflection

What did I learn that I didn't know before?

Before this project, I was familiar with Pandas and tabular data analysis but had limited experience with spatial data. I learned how **GeoDataFrames** combine tabular data with geometry, how **CRS** supports correct spatial operations, and how **spatial joins** connect points with the polygons they fall within how Spatial aggregation helps to do statistical analysis.

What was the most difficult part?

The most difficult part was the spatial join and handling the 16 points that did not match a district polygon. I learned to investigate these cases instead of simply ignoring them and used the original district information as a documented fallback.

Which technical choice am I most satisfied with?

I am most satisfied with using the **spatial join** rather than simply merging district names from the CSV. This made the project a genuine geospatial analysis.

What would I do differently?

If I started the project again, I would plan the final visualizations earlier so that the analysis and the questions being investigated were connected from the beginning.

What is the natural next step?

My next step would be to deepen my knowledge of **GeoPandas** by working with more advanced spatial operations, such as spatial overlays, buffering, spatial aggregation and different types of spatial joins. This would allow me to move from basic geospatial analysis towards more advanced **spatial Data Science** applications.

What grade do I think my work warrants?

I believe my work warrants **VG (Pass with Distinction)**.

The project demonstrates a complete geospatial workflow including data cleaning, GeoDataFrame creation, CRS handling, boundary integration, spatial joining and visualization. I also solved practical implementation problems such as unmatched spatial points cross check with district boundary and inclusion as fallback.